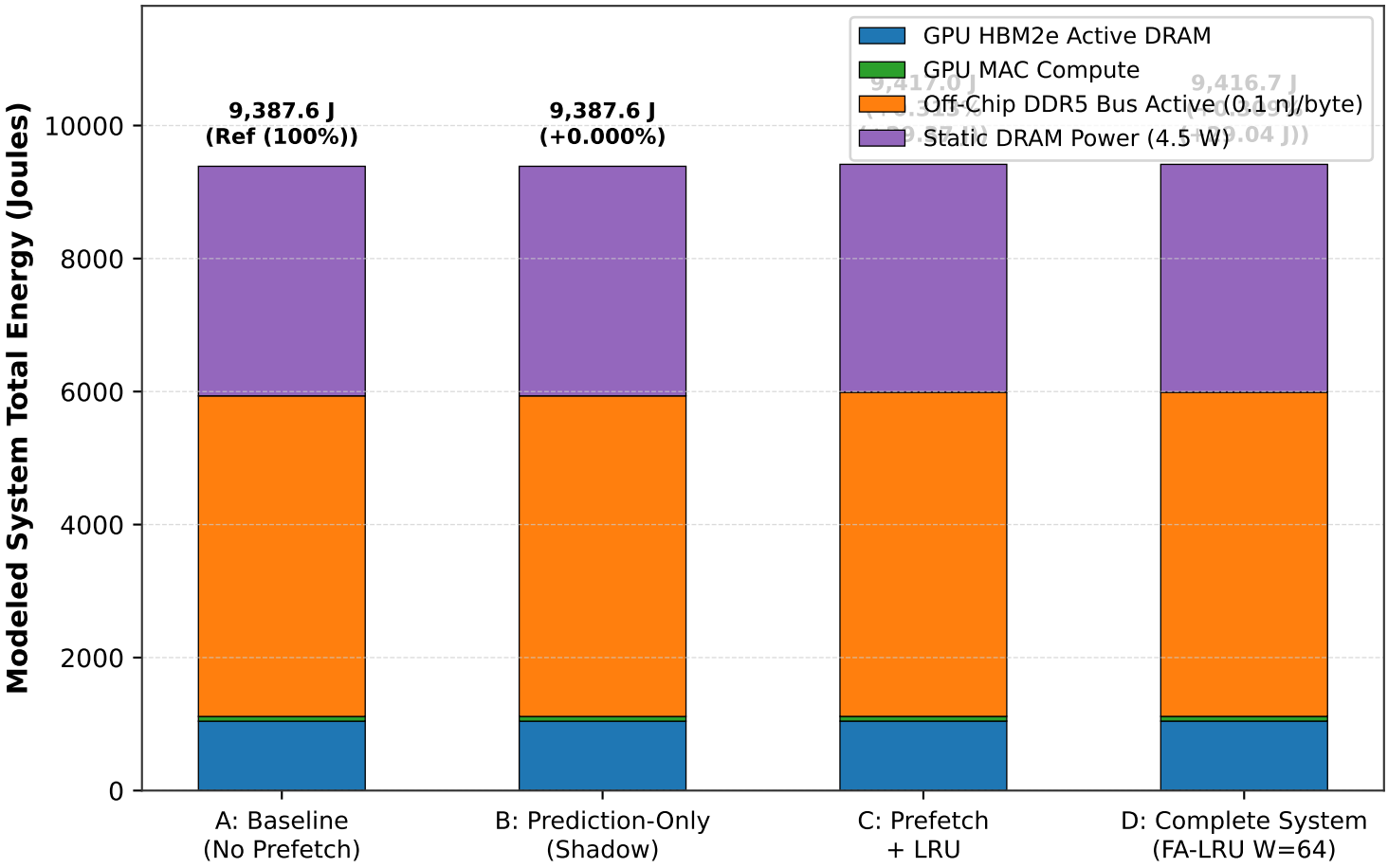
Figure 1: Modeled System Total Energy across Anchor Configurations
(B=32, P=2048, T=128, K=4, H=1)
Modeled System Total Energy (Joules)
0
2000
4000
6000
8000
10000
GPU HBM2e Active DRAM
GPU MAC Compute
Off-Chip DDR5 Bus Active (0.1 nJ/byte)
Static DRAM Power (4.5 W)
9,387.6 J
(Ref (100%))
9,387.6 J
(+0.000%)
9,417.0 J
9,416.7 J
A: Baseline
(No Prefetch)
B: Prediction-Only
(Shadow)
C: Prefetch
+ LRU
D: Complete System
(FA-LRU W=64)